

MH1301 Discrete Mathematics

Tutorial 7

Academic Year 2025/2026, Semester 2

Quantitative Research Society @NTU

August 17, 2026

Overview

This tutorial develops the basic counting and structural theory of finite simple graphs. The main themes are:

- counting labelled simple graphs with prescribed numbers of vertices and edges;
- regular graphs and the Handshaking Lemma;
- classification of small graphs up to isomorphism;
- degree arguments and the pigeonhole principle in graph theory;
- complements of graphs, self-complementary graphs, and invariance under isomorphism;
- counting labelled subgraphs of a complete graph.

Question map.

- Question 1: labelled simple graphs with n vertices and m edges.
- Questions 2–5: regularity, edge counts, and non-isomorphic small graphs.
- Questions 6–9: complements, self-complementary graphs, and cycle graphs.
- Question 10: counting subgraphs of K_3 with labelled vertices.

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Question 1 (Counting simple graphs with n vertices and m edges)

Problem

How many different possible simple graphs of n vertices and m edges are there?

Assume that $G = (V, E)$ with

$$V = \{1, 2, \dots, n\}.$$

Solution

Method 1: Choose the edge set directly

Because the graph is simple, an edge is an unordered pair of distinct vertices. Hence the total number of *possible* edges on the fixed vertex set

$$V = \{1, 2, \dots, n\}$$

is

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

A simple graph with exactly m edges is determined uniquely by choosing which m of these $\binom{n}{2}$ possible edges are present.

Therefore the number of such graphs is

$$\binom{\binom{n}{2}}{m} = \binom{n(n-1)/2}{m}.$$

Hence

$$\boxed{\binom{\binom{n}{2}}{m} = \binom{n(n-1)/2}{m}}.$$

Method 2: Direct counting formulas

$$\#\{G = (V, E) : |V| = n, |E| = m\} = \binom{\binom{n}{2}}{m}.$$

Hence

$$\boxed{\binom{\binom{n}{2}}{m}}.$$

Method 3: Incidence-vector viewpoint

List the $\binom{n}{2}$ possible edges of the complete graph K_n in some fixed order:

$$e_1, e_2, \dots, e_{\binom{n}{2}}.$$

Any simple graph G on the labelled vertex set V can then be encoded by a binary vector

$$(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\binom{n}{2}}),$$

where

$$\varepsilon_i = \begin{cases} 1, & \text{if } e_i \in E(G), \\ 0, & \text{if } e_i \notin E(G). \end{cases}$$

The graph has exactly m edges precisely when this binary vector has exactly m entries equal to 1.

Thus the number of such graphs is exactly the number of binary strings of length $\binom{n}{2}$ with Hamming weight m , which is

$$\binom{\binom{n}{2}}{m}.$$

Therefore

$$\boxed{\binom{\binom{n}{2}}{m} = \binom{n(n-1)/2}{m}}.$$

Question 2 (Regular graphs and graphs with many edges)

Problem

- (a) Do simple, 3-regular graphs with 5 edges exist?
- (b) How many graphs with 6 vertices and 14 edges (with vertex labels $\{1, \dots, 6\}$) are there?

Solution

Method 1: Handshaking Lemma and direct counting

- (a) Suppose such a graph exists, and let it have n vertices. Since it is 3-regular, every vertex has degree 3, so the sum of the degrees is

$$3n.$$

On the other hand, the Handshaking Lemma says that the sum of the degrees equals twice the number of edges:

$$\sum_{v \in V} \deg(v) = 2|E| = 2 \cdot 5 = 10.$$

Therefore

$$3n = 10,$$

which is impossible because n must be an integer.

Hence no such simple graph exists:

No, a simple 3-regular graph with 5 edges does not exist.

- (b) A simple graph on 6 labelled vertices has at most

$$\binom{6}{2} = 15$$

edges. So a graph with 14 edges is obtained by removing exactly one edge from K_6 .

There are exactly 15 possible edges in K_6 , and each choice of the omitted edge yields a distinct labelled graph. Therefore the number of such graphs is

$$15.$$

Hence

$$15.$$

Method 2: Direct use of graph formulas

- (a) If G is 3-regular with 5 edges, then

$$3|V| = 2|E| = 10,$$

impossible. Hence

No such graph exists.

- (b)

$$\#\{G = (V, E) : |V| = 6, |E| = 14\} = \binom{\binom{6}{2}}{14} = \binom{15}{14} = 15.$$

Hence

15.

Method 3: Parity/integrality and the general counting formula

- (a) In a 3-regular graph, every vertex has odd degree. By the Handshaking Lemma, the number of odd-degree vertices must be even. So the number of vertices n must be even.

But if the graph has 5 edges, then

$$3n = 2|E| = 10,$$

so $n = 10/3$, which is not even, and in fact not even an integer. This contradiction shows that such a graph cannot exist.

Therefore

No such graph exists.

- (b) By Question 1, the number of labelled simple graphs on 6 vertices with 14 edges is

$$\binom{\binom{6}{2}}{14} = \binom{15}{14} = 15.$$

So

15.

Question 3 (Non-isomorphic simple graphs with few edges)

Problem

- (a) How many non-isomorphic simple graphs with 3 vertices and 2 edges are there?
- (b) How many non-isomorphic simple graphs with 4 vertices and 2 edges are there?

Solution

Method 1: Direct shape classification

- (a) A simple graph on 3 vertices with 2 edges must use all 3 vertices, because on only 2 vertices there can be at most 1 edge.

So the two edges must share a common vertex, producing a path of length 2:



There is only one such shape up to relabelling. Therefore the number of non-isomorphic graphs is

1.

- (b) Let G be a simple graph on 4 vertices with 2 edges. There are only two possibilities:
- the two edges share a common endpoint, giving a path of length 2 together with one isolated vertex;
 - the two edges are disjoint, giving two disjoint edges.

These two graphs are not isomorphic, because the first has degree multiset

$$\{2, 1, 1, 0\},$$

whereas the second has degree multiset

$$\{1, 1, 1, 1\}.$$

Hence there are exactly

2

non-isomorphic simple graphs with 4 vertices and 2 edges.

Method 2: Classification by degree sequence

- (a) A graph with 3 vertices and 2 edges has total degree

$$2 \cdot 2 = 4.$$

The only possible degree sequence for a simple graph on 3 vertices with total degree 4 is

$$(2, 1, 1).$$

This degree sequence determines exactly the path on 3 vertices. Hence there is only one isomorphism class:

1.

(b) A graph with 4 vertices and 2 edges has total degree

$$2 \cdot 2 = 4.$$

The possible non-increasing degree sequences of length 4 summing to 4 and realizable by simple graphs are:

$$(2, 1, 1, 0), \quad (1, 1, 1, 1).$$

The first corresponds to a path of length 2 together with an isolated vertex. The second corresponds to two disjoint edges.

Since non-isomorphic graphs must have different degree sequences, these yield two distinct isomorphism classes, and there are no others. Therefore

2.

Question 4 (Ex. 10.2.18: Two vertices of the same degree)

Problem

Show that in a simple graph with at least two vertices, there must be two vertices that have the same degree.

Solution

Method 1: Pigeonhole principle with forbidden extreme pair

Let G be a simple graph with $n \geq 2$ vertices. Each vertex degree lies in the set

$$\{0, 1, 2, \dots, n-1\}.$$

However, a simple graph cannot simultaneously have:

- a vertex of degree 0, and
- a vertex of degree $n-1$.

Indeed, if some vertex has degree $n-1$, then it is adjacent to every other vertex, so no vertex can have degree 0.

Therefore the degrees of the n vertices can take values from at most one of the two sets

$$\{0, 1, \dots, n-2\} \quad \text{or} \quad \{1, 2, \dots, n-1\},$$

each of which contains only $n-1$ possible values.

So we are assigning n vertices (the pigeons) to at most $n-1$ possible degree values (the pigeonholes). By the pigeonhole principle, two vertices must have the same degree.

Hence

In every simple graph with at least two vertices, two vertices have the same degree.

Method 2: Contradiction by assuming all degrees are distinct

Suppose, for contradiction, that all n vertices have distinct degrees. Since the possible degrees in a simple graph with n vertices are integers between 0 and $n-1$, the only way to have n distinct degrees is to realize every one of the values

$$0, 1, 2, \dots, n-1.$$

In particular, there would be both:

- a vertex of degree 0, and
- a vertex of degree $n-1$.

But a vertex of degree $n - 1$ is adjacent to every other vertex, including the alleged degree-0 vertex. That contradicts the fact that the degree-0 vertex is adjacent to no one.

Therefore our assumption was false, and at least two vertices must have equal degree. Thus

At least two vertices have the same degree.

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Question 5 (Non-isomorphic simple graphs with 4 vertices and 3 edges)

Problem

How many non-isomorphic simple graphs with 4 vertices and 3 edges are there?

Solution

Method 1: Classification by degree sequence

Let G be a simple graph with 4 vertices and 3 edges. Then the sum of its degrees is

$$2|E| = 2 \cdot 3 = 6.$$

We list the possible degree sequences of a simple graph on 4 vertices whose entries are nonnegative integers, at most 3, and sum to 6. The realizable possibilities are:

$$(3, 1, 1, 1), \quad (2, 2, 2, 0), \quad (2, 2, 1, 1).$$

These correspond respectively to:

- the star $K_{1,3}$;
- a triangle K_3 together with one isolated vertex;
- the path P_4 on 4 vertices.

These three graphs are pairwise non-isomorphic because isomorphic graphs must have the same degree sequence, and these degree sequences are different.

Therefore the number of non-isomorphic simple graphs with 4 vertices and 3 edges is

$$\boxed{3}.$$

Method 2: Case split according to whether a triangle is present

Let G be a simple graph on 4 vertices with 3 edges.

Case 1: G contains a triangle. A triangle already uses 3 edges, so the fourth vertex must be isolated. Thus G is isomorphic to

$$K_3 \cup K_1.$$

Case 2: G contains no triangle. Since G has 4 vertices and 3 edges and no triangle, G is acyclic; otherwise it would contain a cycle, and with only 4 vertices and 3 edges the only possible cycle would be a triangle.

So G is a forest with 4 vertices and 3 edges. Any forest on n vertices with $n - 1$ edges is a tree, so here G must be a tree on 4 vertices. Up to isomorphism, there are exactly two trees on 4 vertices:

$$P_4 \quad \text{and} \quad K_{1,3}.$$

Thus the three possibilities are:

$$K_3 \cup K_1, \quad P_4, \quad K_{1,3}.$$

These are non-isomorphic, for example because their degree sequences are different:

$$(2, 2, 2, 0), \quad (2, 2, 1, 1), \quad (3, 1, 1, 1).$$

Therefore

$$\boxed{3.}$$

Question 6 (Ex. 10.2.60 and Ex. 10.2.65: Complements of graphs)

Problem

The complement of a graph $G = (V, E)$ is the graph $\overline{G} = (V, \overline{E})$ where $\overline{E}$ contains exactly those edges that are not in E .

- (a) Show that if G is a simple graph with n vertices, then the union of G and $\overline{G}$ is K_n .
- (b) If G is a simple graph with 15 edges and $\overline{G}$ has 13 edges, how many vertices does G have?

Solution

Method 1: Edge-set partition argument

- (a) Let $G = (V, E)$ be a simple graph with $|V| = n$. By definition, the complement $\overline{G}$ has the same vertex set V , and its edge set $\overline{E}$ consists of exactly those unordered pairs of distinct vertices that are *not* in E .

Thus for every pair of distinct vertices $\{u, v\} \subseteq V$, exactly one of the following holds:

$$\{u, v\} \in E \quad \text{or} \quad \{u, v\} \in \overline{E}.$$

Therefore the union of the edge sets of G and $\overline{G}$ contains *every* possible edge on the vertex set V . But the simple graph on V containing all possible edges is precisely the complete graph K_n .

Hence

$$\boxed{G \cup \overline{G} = K_n.}$$

- (b) Since G and $\overline{G}$ partition the edges of K_n , we have

$$|E(G)| + |E(\overline{G})| = |E(K_n)| = \binom{n}{2}.$$

Given

$$|E(G)| = 15, \quad |E(\overline{G})| = 13,$$

we obtain

$$\binom{n}{2} = 15 + 13 = 28.$$

So

$$\frac{n(n-1)}{2} = 28 \implies n(n-1) = 56.$$

Hence

$$n^2 - n - 56 = 0 = (n-8)(n+7).$$

Since $n > 0$, we get

$$n = 8.$$

Therefore

$$\boxed{8.}$$

Method 2: Adjacency-based reasoning

- (a) In a simple graph with vertex set V , the complete graph K_n consists of all unordered pairs of distinct vertices.

For any two distinct vertices $u, v \in V$, if u and v are adjacent in G , then the edge $\{u, v\}$ belongs to G . If they are not adjacent in G , then by the definition of complement, they are adjacent in $\overline{G}$. Thus every pair of distinct vertices is adjacent in at least one of the graphs G or $\overline{G}$.

So the union graph has all possible adjacencies, which means it is K_n . Therefore

$$\boxed{G \cup \overline{G} = K_n.}$$

- (b) Since $G \cup \overline{G} = K_n$, the total number of edges across G and $\overline{G}$ is the number of edges in K_n , namely

$$\binom{n}{2}.$$

We are told that

$$15 + 13 = 28 = \binom{n}{2}.$$

Now

$$\binom{8}{2} = \frac{8 \cdot 7}{2} = 28,$$

so

$$n = 8.$$

Hence

$$\boxed{8.}$$

Question 7 (Complements preserve isomorphism)

Problem

Suppose that G and H are isomorphic simple graphs. Show that their complementary graphs $\overline{G}$ and $\overline{H}$ are also isomorphic.

Solution

Method 1: Use the same isomorphism directly

Let

$$G = (V_G, E_G), \quad H = (V_H, E_H),$$

and suppose $G \cong H$. Then there exists a bijection

$$f : V_G \rightarrow V_H$$

such that for all distinct vertices $u, v \in V_G$,

$$\{u, v\} \in E_G \iff \{f(u), f(v)\} \in E_H.$$

We must show that the same bijection f is an isomorphism from $\overline{G}$ to $\overline{H}$.

Now

$$\{u, v\} \in E(\overline{G}) \iff \{u, v\} \notin E_G.$$

Using the fact that f preserves adjacency in G and H , we obtain

$$\{u, v\} \notin E_G \iff \{f(u), f(v)\} \notin E_H.$$

But the right-hand side is exactly

$$\{f(u), f(v)\} \in E(\overline{H}).$$

Therefore

$$\{u, v\} \in E(\overline{G}) \iff \{f(u), f(v)\} \in E(\overline{H}).$$

So f is an isomorphism $\overline{G} \rightarrow \overline{H}$, and hence

$$\boxed{\overline{G} \cong \overline{H}.}$$

Method 2: Adjacency-matrix proof

Suppose G and H are isomorphic simple graphs on n vertices. Then there exists a permutation matrix P such that their adjacency matrices satisfy

$$A_H = P^T A_G P.$$

For a simple graph on n vertices, the adjacency matrix of its complement is

$$A_{\overline{G}} = J - I - A_G,$$

where J is the $n \times n$ all-ones matrix and I is the identity matrix.

Similarly,

$$A_{\overline{H}} = J - I - A_H.$$

Substitute $A_H = P^T A_G P$:

$$A_{\overline{H}} = J - I - P^T A_G P.$$

Because P is a permutation matrix,

$$P^T J P = J, \quad P^T I P = I.$$

Hence

$$\begin{aligned} A_{\overline{H}} &= P^T J P - P^T I P - P^T A_G P \\ &= P^T (J - I - A_G) P \\ &= P^T A_{\overline{G}} P. \end{aligned}$$

Therefore the adjacency matrices of $\overline{G}$ and $\overline{H}$ are permutation-similar, so the graphs are isomorphic:

$$\boxed{\overline{G} \cong \overline{H}.}$$

Question 8 (Self-complementary graphs on 2, 3, 4 vertices)

Problem

A simple graph G is called *self-complementary* if G and $\overline{G}$ are isomorphic.

- (a) Show that there are no self-complementary simple graphs with two nor three vertices.
- (b) Find a self-complementary simple graph with four vertices.

Solution

Method 1: Edge-count argument plus explicit example

- (a) If a graph G is self-complementary, then $G \cong \overline{G}$. Isomorphic graphs have the same number of edges, so

$$|E(G)| = |E(\overline{G})|.$$

But G and $\overline{G}$ together partition the edges of the complete graph K_n . Hence

$$|E(G)| + |E(\overline{G})| = \binom{n}{2}.$$

So for a self-complementary graph,

$$2|E(G)| = \binom{n}{2}.$$

Thus $\binom{n}{2}$ must be even.

Now:

$$\binom{2}{2} = 1 \quad \text{is odd}, \quad \binom{3}{2} = 3 \quad \text{is odd}.$$

Therefore no self-complementary simple graph can exist on 2 or on 3 vertices.

Hence

There are no self-complementary simple graphs with 2 or 3 vertices.

- (b) On 4 vertices, $\binom{4}{2} = 6$, so a self-complementary graph must have exactly

$$\frac{6}{2} = 3$$

edges.

Take the path P_4 on vertices 1, 2, 3, 4:

$$E(G) = \{\{1, 2\}, \{2, 3\}, \{3, 4\}\}.$$

Its complement has edge set

$$E(\overline{G}) = \{\{1, 3\}, \{1, 4\}, \{2, 4\}\},$$

which is again a path on 4 vertices, namely

$$3 - 1 - 4 - 2.$$

So $G \cong \overline{G}$. For example, the bijection

$$f(1) = 3, \quad f(2) = 1, \quad f(3) = 4, \quad f(4) = 2$$

maps the edges of G to the edges of $\overline{G}$.

Therefore one self-complementary graph on 4 vertices is P_4 , and we may state:

A self-complementary graph on 4 vertices is the path P_4 .

Method 2: Explicit classification of small graphs

(a) For $n = 2$, there are only two simple graphs up to isomorphism:

- the empty graph with 0 edges;
- the single-edge graph with 1 edge.

These are complements of each other, but they are not isomorphic because they have different numbers of edges.

For $n = 3$, the isomorphism classes are:

- the empty graph (0 edges);
- one edge plus an isolated vertex (1 edge);
- a path of length 2 (2 edges);
- the triangle K_3 (3 edges).

Complements pair them as

$$0 \leftrightarrow 3, \quad 1 \leftrightarrow 2.$$

In each pair, the two graphs have different degree sequences, so they are not isomorphic. Therefore there are no self-complementary graphs on 2 or 3 vertices.

Thus

No self-complementary simple graphs exist for $n = 2$ or $n = 3$.

(b) By Question 5, up to isomorphism the simple graphs on 4 vertices with 3 edges are:

$$K_{1,3}, \quad K_3 \cup K_1, \quad P_4.$$

Now

- $\overline{K_{1,3}} \cong K_3 \cup K_1$;
- $\overline{K_3 \cup K_1} \cong K_{1,3}$;
- $\overline{P_4} \cong P_4$.

So the only self-complementary isomorphism class is P_4 .

Hence

A self-complementary graph with 4 vertices is P_4 .

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Question 9 (Self-complementary graphs with five vertices and cycles)

Problem

- (a) Find a self-complementary simple graph with five vertices.
 (b) For which integers n is C_n self-complementary?

Solution

Method 1: Use C_5 explicitly

- (a) Let

$$G = C_5$$

with vertex set

$$V = \{1, 2, 3, 4, 5\}$$

and edge set

$$E(G) = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 5\}, \{5, 1\}\}.$$

The complement has the remaining 5 edges:

$$E(\overline{G}) = \{\{1, 3\}, \{1, 4\}, \{2, 4\}, \{2, 5\}, \{3, 5\}\}.$$

These edges form the cycle

$$1 - 3 - 5 - 2 - 4 - 1.$$

Hence $\overline{G} \cong C_5 \cong G$, so C_5 is self-complementary.

Thus an example is

$$\boxed{C_5}.$$

- (b) If C_n is self-complementary, then $C_n \cong \overline{C_n}$. Isomorphic graphs have the same number of edges.

Now C_n has exactly

$$n$$

edges. The complete graph K_n has

$$\binom{n}{2}$$

edges, so $\overline{C_n}$ has

$$\binom{n}{2} - n$$

edges. For self-complementarity we require

$$n = \binom{n}{2} - n.$$

Hence

$$2n = \binom{n}{2} = \frac{n(n-1)}{2}.$$

Multiply by 2:

$$4n = n(n-1).$$

Since $n \geq 3$ for a cycle graph, divide by n :

$$4 = n - 1,$$

so

$$n = 5.$$

Therefore

$$\boxed{C_n \text{ is self-complementary exactly when } n = 5.}$$

Method 2: Degree argument

- (a) Another way to verify that C_5 is self-complementary is to note that C_5 is 2-regular on 5 vertices. In its complement, each vertex is adjacent to all but itself and its two original neighbors, so the complement is also 2-regular on 5 vertices. The only connected 2-regular simple graph on 5 vertices is C_5 , so the complement is again a 5-cycle. Hence

$$\boxed{C_5 \text{ is self-complementary.}}$$

- (b) The cycle C_n is 2-regular. Its complement $\overline{C_n}$ is $(n-3)$ -regular, because each vertex was adjacent to 2 vertices in C_n and to the remaining $n-1-2 = n-3$ vertices in the complement.

If C_n is self-complementary, then $C_n \cong \overline{C_n}$, so isomorphic graphs must have the same degree sequence. Therefore

$$2 = n - 3.$$

Hence

$$n = 5.$$

Thus

$$\boxed{C_n \text{ is self-complementary if and only if } n = 5.}$$

Question 10 (Subgraphs of K_3 with at least one vertex)

Problem

How many subgraphs with at least one vertex does K_3 have? Assume vertices are labelled and distinct.

Solution

Method 1: Count by the number of chosen vertices

Since the vertices are labelled, distinct choices of vertex sets give distinct subgraphs.

Case 1: The subgraph has exactly 1 vertex. Choose the vertex in

$$\binom{3}{1} = 3$$

ways. A graph on one vertex has no edges, so this contributes

$$3.$$

Case 2: The subgraph has exactly 2 vertices. Choose the vertex set in

$$\binom{3}{2} = 3$$

ways. On those two labelled vertices, there is exactly one possible edge, which may be either present or absent. Therefore each such vertex set yields

$$2$$

subgraphs, for a contribution of

$$3 \cdot 2 = 6.$$

Case 3: The subgraph has all 3 vertices. There is

$$\binom{3}{3} = 1$$

choice of vertex set. Among these three vertices there are exactly three possible edges, and each can be chosen independently to be present or absent. Therefore there are

$$2^3 = 8$$

such subgraphs.

Summing all cases:

$$3 + 6 + 8 = 17.$$

Hence

$$\boxed{17.}$$

Method 2: Sum over vertex subsets

A subgraph of K_3 is determined by:

- a nonempty vertex subset $S \subseteq V(K_3)$, and
- an arbitrary subset of the edges among the vertices of S .

If $|S| = k$, then there are

$$\binom{k}{2}$$

possible edges on S , so the number of graphs on that fixed labelled vertex set is

$$2^{\binom{k}{2}}.$$

Hence the total number of nonempty subgraphs is

$$\sum_{k=1}^3 \binom{3}{k} 2^{\binom{k}{2}}.$$

Compute:

$$\binom{3}{1} 2^{\binom{1}{2}} = 3 \cdot 2^0 = 3,$$

$$\binom{3}{2} 2^{\binom{2}{2}} = 3 \cdot 2^1 = 6,$$

$$\binom{3}{3} 2^{\binom{3}{2}} = 1 \cdot 2^3 = 8.$$

Therefore

$$\sum_{k=1}^3 \binom{3}{k} 2^{\binom{k}{2}} = 3 + 6 + 8 = 17.$$

So

$$\boxed{17.}$$